Project Report

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Subject Name: Internet of Things Lab **Subject Code: 24CAP-751**

Aim: IoT-based Laser Security System that detects intrusions using a laser beam and sends real-time alerts via the Blynk cloud.

Objectives:

1. To detect intruders using laser and LDR sensor.
2. To send real-time alerts using Blynk IoT platform.
3. To provide visual and sound alerts using LEDs and buzzer.
4. To display system status on OLED screen.
5. To enable remote monitoring and control of the system.

Components Required:

Sno	Name of Component	Qty.
1.	NodeMCU (ESP8266)	1
2.	Laser Module	1
3.	LDR Sensor	1
4.	OLED Display (128x64)	1
5.	Buzzer	1
6.	LEDs (Red & Green)	2
7.	Breadboard & Wires	1

Details of Components

1. NodeMCU:

The NodeMCU board is a low-cost, open-source IoT platform widely recognized for its user-friendly approach to building Internet-connected projects. Based on the ESP8266 WiFi chip, it combines a microcontroller with WiFi capabilities, enabling developers to create connected devices with ease.

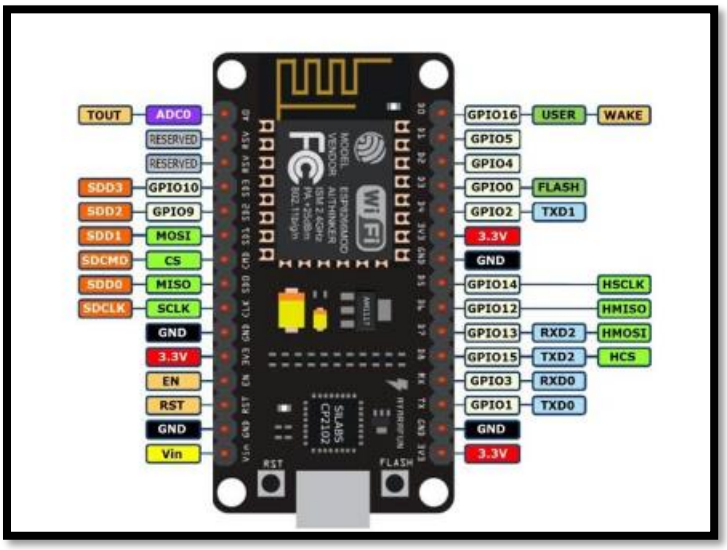

Figure 1 : NodeMCU Board

2. Laser Module:

The laser module emits a focused beam of light used for intrusion detection.
When the beam is interrupted, the system detects it as a security breach and triggers an alert.

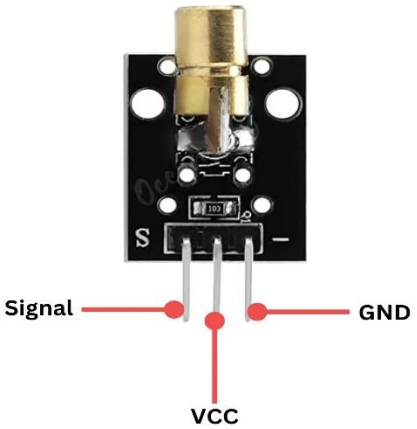

Figure 2: Laser Module

3. LDR Sensor:

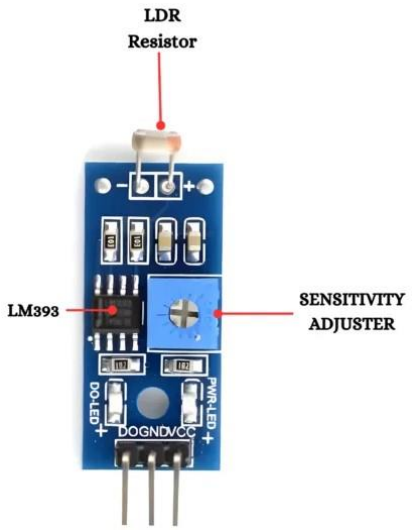

Figure 3:LDR Sensor

The LDR (Light Dependent Resistor) detects the intensity of light falling on it.
When the laser beam is interrupted, the change in light intensity is detected and used to trigger the alarm system.

4. OLED Display:

The OLED display is used to show real-time system status such as SAFE, ALERT, and OFF.
It provides a clear visual interface for monitoring alerts and system activity.

Figure 4: OLED Display

5. Buzzer:

The buzzer is used to produce an audible alarm when an intrusion is detected.
It alerts users immediately by generating a beeping sound during the ALERT condition.

Figure 5:Active Buzzer

6. LEDs (Green & Red):

LEDs are used as visual indicators of the system status.
The green LED indicates SAFE condition, while the red LED indicates ALERT when an intrusion is detected.

Figure 6: LEDs

7. Breadboard & Wires:

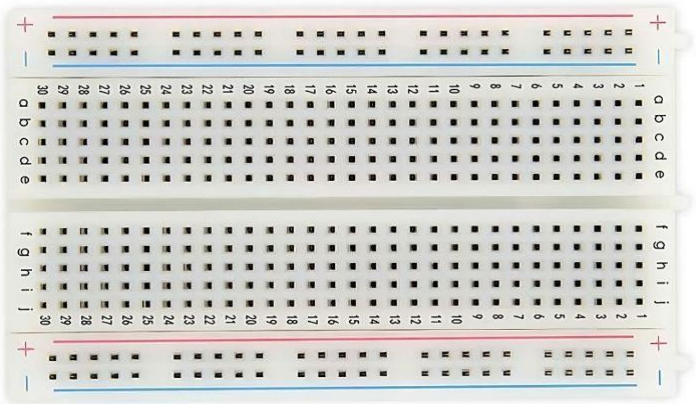

Figure 7: Breadboard

The breadboard is used to assemble the circuit without soldering, making prototyping easy. Connecting wires are used to establish electrical connections between all components.

Block Diagram of Designed Model:

- The NodeMCU (ESP8266) acts as the central controller of the system. The laser module continuously emits a beam onto the LDR sensor. When the beam is interrupted, the NodeMCU detects the change and activates the alarm system.

- The output devices such as the buzzer, LEDs, and OLED display provide real-time alerts and system status. Additionally, the NodeMCU connects to the Blynk cloud via WiFi, allowing the user to monitor and control the system remotely through a mobile application.

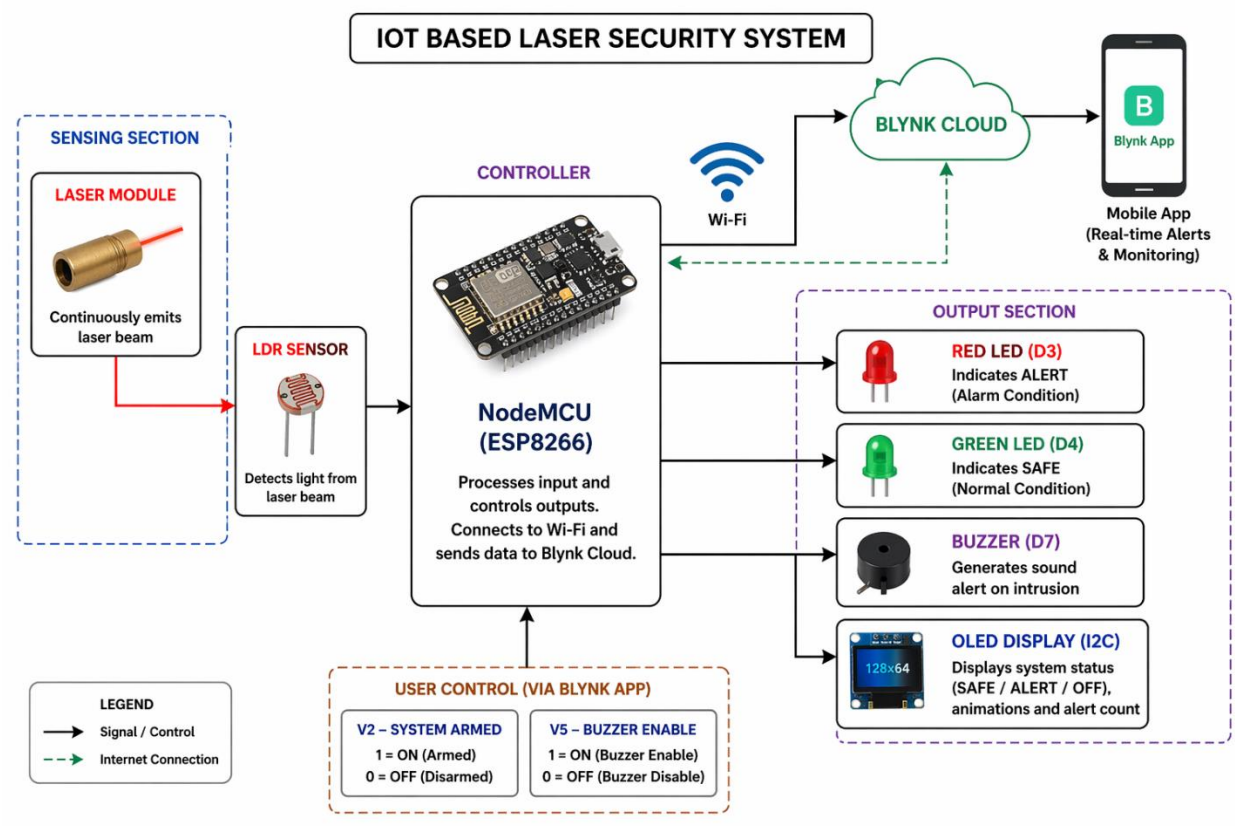

Figure 8: Block Diagram of Model

Working of Designed Model:

- The system uses a laser beam focused on an LDR sensor to continuously monitor the area.
- The NodeMCU reads the LDR signal; when the laser beam is uninterrupted, the system remains in **SAFE mode**.
- If an intruder crosses the path and breaks the laser beam, the LDR detects a change in light intensity.
- To avoid false alarms, the system checks multiple readings (stable count logic) before confirming intrusion.
- Once intrusion is confirmed:

- The **red LED starts blinking**
- The **buzzer activates** (if enabled)

- The **OLED display shows "ALERT"**
- A **notification is sent to the Blynk app** with location details

- The system also maintains an **alert count** for tracking multiple intrusions.
- When the laser beam is restored, the system returns to **SAFE mode**, where the green LED turns ON and the buzzer stops.
- The user can **arm/disarm the system** and control the buzzer remotely using the Blynk application.

Pictures of Prototype:

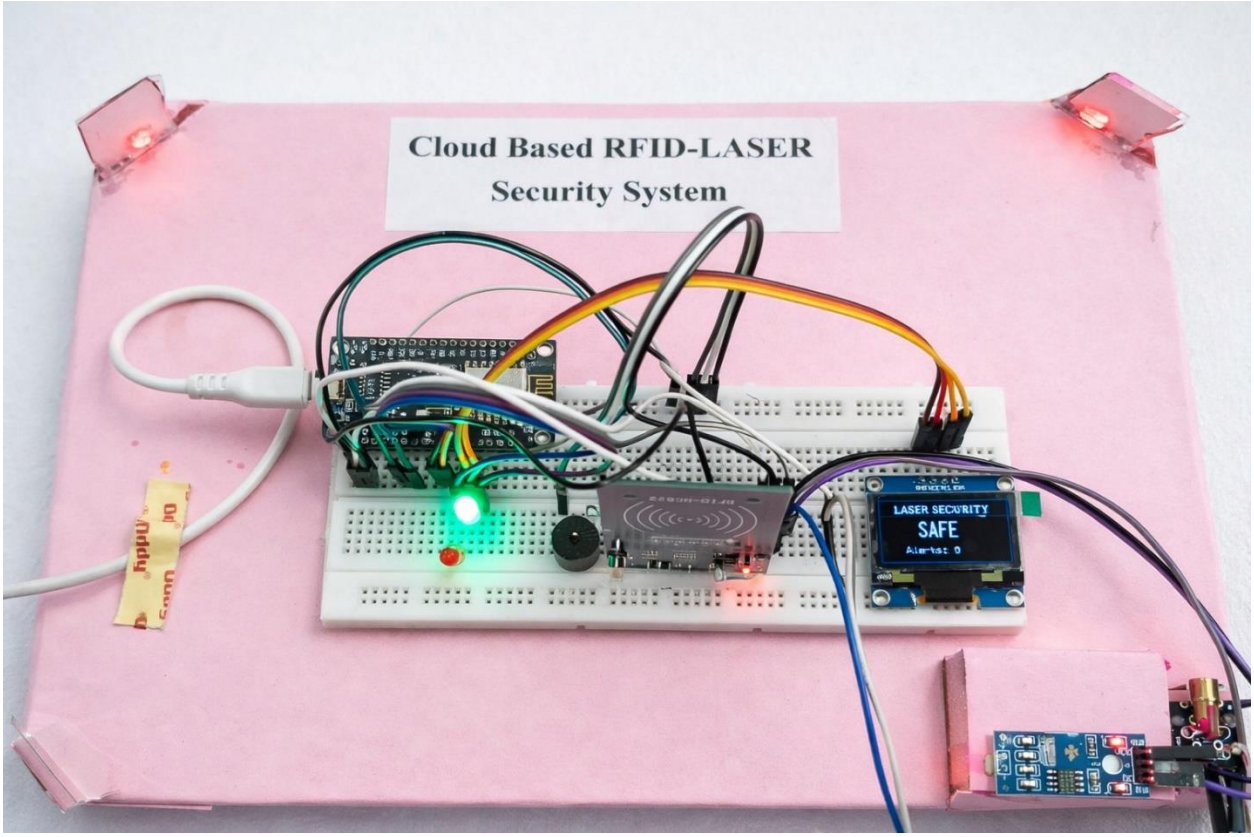

Figure 9: Designed Model

Output of Deigned Model/Prototype

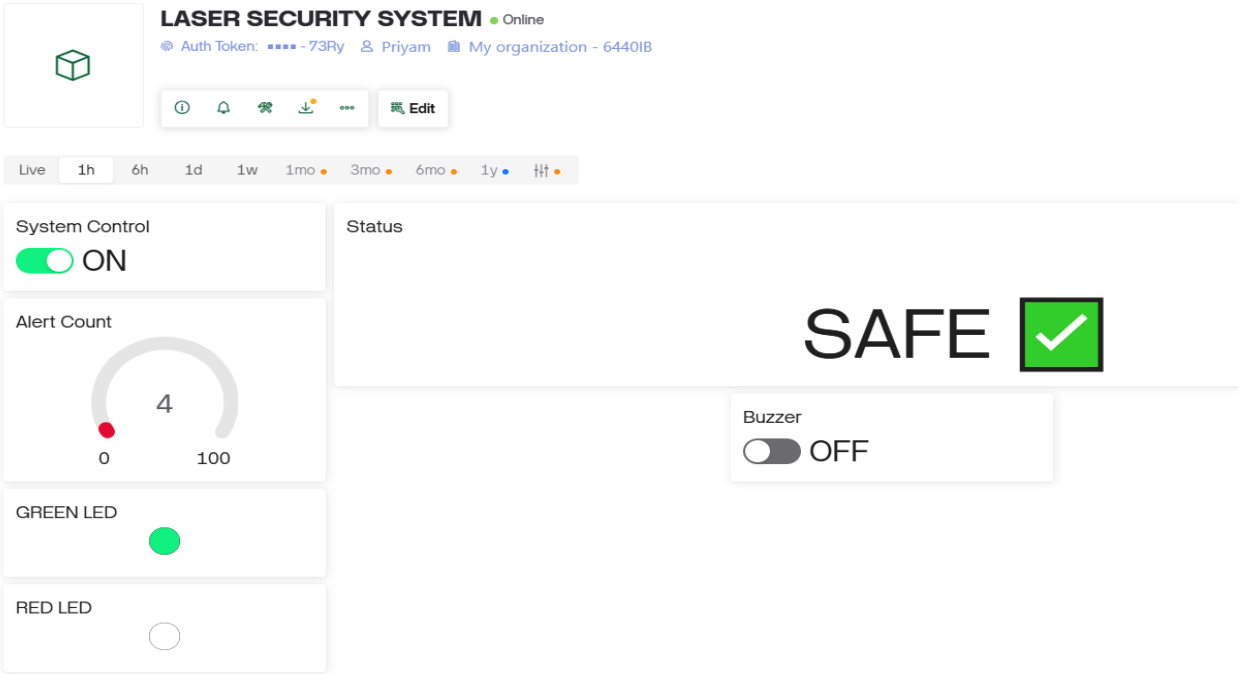

Figure 10: Output on BLYNK Cloud

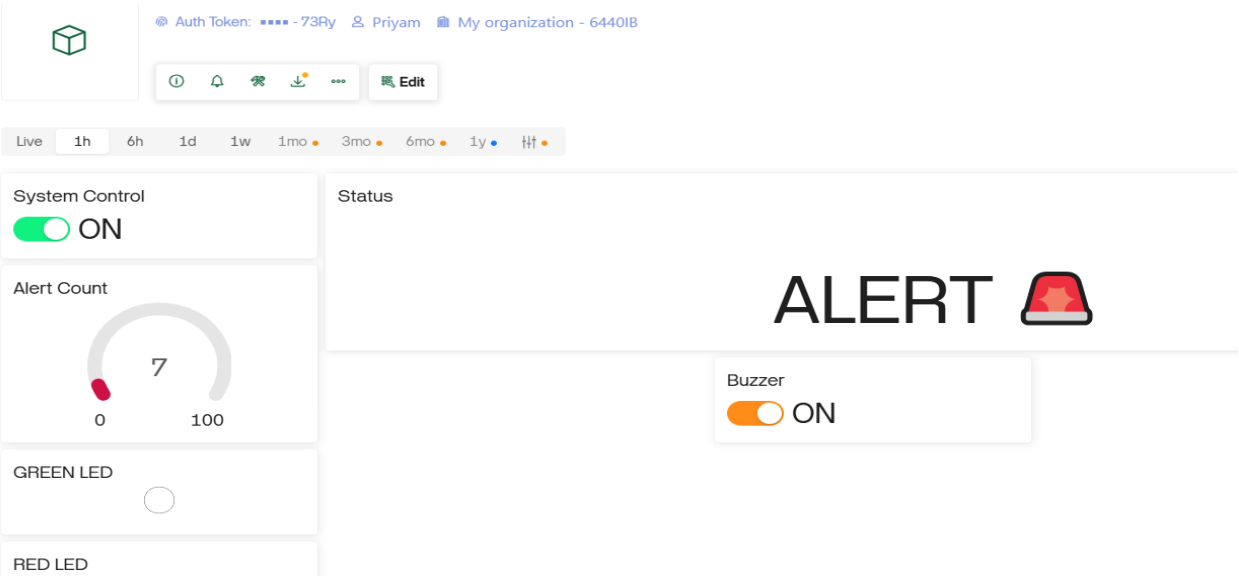

Figure 11 Output on BLYNK Cloud

<u>Learning outcomes</u> (What I have learnt):

1. Learned how to design and implement an IoT-based security system using NodeMCU (ESP8266).

2. Gained practical knowledge of sensor integration (LDR) for intrusion detection.

3. Understood how to use the Blynk platform for real-time monitoring and notifications.

4. Learned to interface output devices like LEDs, buzzer, and OLED display with a microcontroller.

5. Improved skills in embedded programming, circuit design, and system troubleshooting.